Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat
Department of Computer Science and Engineering

B.Tech. II (CSE) Semester-III
Database Management Systems (CS203)

Academic Year: 2026-27

Practical Assignment – 11

Date: 19th and 26th October 2026
(2 Weeks)

For your registered project:

A) Write SQL statements for the following, execute from SQL Window and **NOTE DOWN THE EXECUTION TIME** for each query:

1) To display all the data from one of the main tables.
2) To insert data into various tables maintaining integrity constraints.
3) To display ONE AGGREGATE FUNCTION RESULT from one of the main tables.
4) To display the data from one of the main tables using condition.
5) To display JOIN OPERATIONS on any two tables.
6) To display some records using SUBQUERY from more than one different table.
7) To display some records using CORELATEDQUERY from more than one different table.
8) To update any one table data.

B) Create appropriate 2 or more indices on your tables for above queries of SECTION A, **NOTE DOWN THE EXECUTION TIME** for each query after the index and analyze the impact of index on each query.

C) CREATE VIEW from two tables to combined on some condition like one city data, or one age-group data.

1) Execute your saved procedure to insert data/to display data from view.
2) Execute your trigger for some validation on the view, to analyze the impact of trigger on view.

D) Prepare FRONTEND using HTML and node.js or any other technology:

1) Provide UI feature: Menu to perform the various operations like Insert, display, update, delete.
2) For the display result of query no. 1 of Section A, obtain the data in frontend and display data in HTML Table.
3) With appropriate form feature(s), to get the data for your insertion query no.2 of Section A and insert into your DBMS table.
4) For the query no. 3 of Section A, provide the drop down box for different aggregation function, execute the sql query in DBMS and display the result in HTML page.
5) For the query no. 4 of Section A, provide the drop down box for different conditions and operators, execute the sql query in DBMS and display the result in HTML page.
6) For the query no. 6 and 7 of Section A, provide the drop down box for different condition options to get data from multiple tables, execute the sql query in DBMS and display the result in HTML

page.

7) For the query no. 8 of Section A, provide the form controls to get the data to update in the DBMS table, execute the sql query in DBMS and display the updated result in HTML
8) For the query no. 5 of Section A, display the 2 dropdown boxes with table names of your project, provide the one more dropdown box with different join operations, and upon submit show the result in front end.

E) **Create dashboard of your project** in front end where statistics of your all tables with proper message is displayed.

F) ONLY for Query Nos. 1, 3, and 4, **display data in XML format** in front end.